

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed, built, and tested **Spring Boot REST APIs** and microservices for enterprise Java applications, implementing core **data structures** and **algorithms** to optimize query performance by **40%**; collaborated with cross-functional product teams to deliver quality end-user features balancing innovation with production reliability.
- Developed **Python** automation scripts and ETL pipelines for data transformation workflows, applying **software design** principles to handle **500K+ daily records**; identified and designed solutions for complex technical challenges in data processing, supported users maintaining deployment infrastructure across AWS and on-premise environments.
- Engineered **AWS cloud-native** features leveraging **EC2, S3, Lambda, and CloudWatch** for serverless event processing; designed and tested distributed systems handling **10M+ monthly transactions**, balancing delivery with R&D; through iterative feature development and performance optimization.
- Led design and implementation of production monitoring dashboards using **Splunk and Dynatrace**, architecting real-time alerting systems for **P1/P2 incident response** with **99.9% SLA uptime**; solved complex problems with context—collaborating with operations and engineering teams to diagnose root causes and validate fixes in production-grade database deployments.
- Demonstrated mastery of **software design fundamentals** through **CI/CD pipeline development** with **Jenkins and Git**, implementing automated testing strategies that reduced defect escape rate by **35%**; maintained code quality through peer reviews and contributed to architectural decisions balancing system reliability with feature velocity.
- Applied **computer science theory**—data structures, algorithms, relational database optimization—to solve real-world problems in **PostgreSQL, MySQL, and Oracle**; wrote and tested complex SQL queries and schema migrations supporting **enterprise-scale systems**, ensuring data integrity across interconnected business applications serving thousands of concurrent users.

SKILLS

Programming Languages – Python, Java, C++, JavaScript, Go, Ruby, Node.js.

Software Design & CS Fundamentals – Data structures, algorithms, software architecture, design patterns, REST API design, microservices.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, CI/CD (Jenkins).

Databases & Data Engineering – PostgreSQL, MySQL, Oracle, SQL optimization, schema design, ETL, data modeling, data warehousing.

Monitoring & Reliability – Splunk, Dynatrace, CloudWatch, SLA management, incident response, production support.

Testing & Quality Assurance – Unit testing, integration testing, automated testing frameworks, defect tracking (Jira), code review.

Collaboration & Agile – Cross-functional team coordination, Scrum ceremonies, Agile development, Confluence documentation, requirements gathering.

EDUCATION AND TRAINING

PROJECTS

Distributed Database Feature Engine & Query Optimization Platform (HCL Technologies)

- Designed and built a **Java Spring Boot**-based feature platform that enabled MongoDB-like query optimization for **3 enterprise clients**, implementing advanced **data structures** (B-trees, hash indices) and **algorithms** (cost-based query planning) to accelerate complex queries by **60%** across **15TB+ datasets**.
- Architected **Python** and **Go** microservices for distributed feature flagging and canary deployment, integrating with **AWS Lambda, CloudWatch, and S3** to manage staged rollouts; tested features end-to-end using automated test suites that achieved **92% code coverage**, catching **18 critical regressions** before production.
- Led cross-functional design workshops with product managers and platform engineers to translate user requirements into technical specifications for **3 major features**; balanced delivery velocity with R&D; by prototyping **2 experimental indexing strategies**, reducing time-to-MVP by **8 weeks** while maintaining production stability.
- Implemented comprehensive monitoring and observability using **Dynatrace and CloudWatch**, building dashboards to track query latency, throughput (**50K queries/sec peak**), and resource utilization; designed **automated incident response** playbooks reducing MTTR from **45 minutes to 8 minutes** for deployment-related issues.
- Collaborated with database operations teams and enterprise support to design **user-facing documentation, training materials, and onboarding guides**; supported customers maintaining production MongoDB-compatible deployments through technical consultation, resulting in **98% feature adoption rate** within **6 months**.

Real-Time Payment Messaging & SLA Monitoring Dashboard (HCL Technologies)

- Designed and tested a **Python + Flask backend** and **React frontend** dashboard for real-time payment transaction visibility, processing **2M+ SWIFT MT/MX messages daily**; implemented **data structures** for efficient transaction state tracking and **algorithms** for SLA breach prediction, achieving **99.95% uptime** for a Tier-1 banking client.
- Built automated **ETL pipelines** integrating **PostgreSQL and Oracle databases** with **AWS CloudWatch** for cross-border payments monitoring; designed **SQL queries and stored procedures** for ISO 20022 message reconciliation and SEPA compliance validation, reducing manual audit overhead by **70%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)